

Rotavirus Infection in Children <5 Years of Age in Suzhou, China, 2013–2019

Disease Burden, Genotype Distribution and Seasonality

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Background: This study aimed to determine the disease burden and strain distribution of rotavirus in children with diarrhea <5 years old in Suzhou, China.

Methods: The study was conducted among children with diarrhea <5 years old at Suzhou University Affiliated Children's Hospital from 2013 to 2019. Rotavirus antigen was detected in clinical laboratory and then sent to Suzhou Centers for Disease Control and Prevention for further molecular analysis. Group A rotavirus (RVA) was detected through enzyme-linked immunosorbent assays, and G-genotype and P-genotype of RVA were tested using reverse transcription-polymerase chain reaction.

Results: Of a total of 198,130 children with diarrhea, 70,813 (35.7%) were positive for RVA; RVA-related diarrhea was detected in 7798 (20.7%, n = 7798/37,710) inpatients and 63,015 (39.3%, n = 63,015/160,420) outpatients. Most children (92.0%, n = 65,171/70,813) positive for RVA were found as children <3 years old. Children 12–35 months old were reported as the highest prevalence among all age groups. The seasonal peak of RVA was in the autumn and winter. Among all 673 RVA strains genotyped, the G9P[8] strain was reported to be persistently predominant in the pediatric population from 2013 to 2019.

Conclusions: The burden of diarrhea disease due to rotavirus infection remains high in Suzhou.

Key Words: rotavirus infection, childhood diarrhea, prevalence, genotypes (*Pediatr Infect Dis J* 2022;41:375–380)

Diarrhea has been found as the second leading cause of death in children <5 years of age, causing approximately 500,000 deaths annually worldwide.^{1,2} In such an age group of children, rotavirus

(RV) has been reported as the single most common cause of severe diarrhea, which is estimated at 10 million severe cases and 118,000 to 183,000 deaths each year.^{1–3} Despite the decreasing number of deaths due to diarrhea, RV is still the major global cause of diarrhea-related morbidity and mortality in children <5 years of age.⁴ Group A rotavirus (RVA) is known as the most important pathogen that causes infective diarrhea in infants.⁵ RVAs fall into G and P types in accordance with the sequence diversity of the genes encoding the outer viral proteins VP7 (glycoprotein) and VP4 (protease-sensitive protein).^{6,7} From the global perspective, 5 strains, that is, G1P[8], G2P[4], G3P[8], G4P[8] and G9P[8], have caused most of the severe diarrhea illness in children in most nations over the past few years.⁸

Most of RV deaths are reported in Asia and sub-Saharan Africa.^{3,9} In 2013, World Health Organization (WHO) Position Paper on Rotavirus Vaccines recommended that RV vaccines should be covered in all national immunization programs and considered a priority, particularly in nations with high RV gastroenteritis-related fatality rates (eg, south and south-eastern Asia and sub-Saharan Africa).¹⁰ Currently, there are 4 WHO prequalified RV vaccines. Besides the globally available RotaTeq (Merck & Co., Inc, West Point, PA, USA) and Rotarix (GlaxoSmithKline Biologicals SA, Rixensart, Belgium), Rotavac (Bharat Biotech, Hyderabad, India) and Rotasiil (Serum Institute of India, Pune, Maharashtra, India) have been licensed and manufactured in India, and prequalified by WHO in 2018.^{9,11} RV vaccines were introduced in 108 countries by the end of 2019, and global coverage was estimated at 39%.¹² In China, one of the high-burden nations for diarrhea,¹³ however, RV vaccination has not been included into the national immunization program thus far. RotaTeq, one of international RV vaccines, was licensed for use in China in the middle of 2018. Moreover, there was a domestic vaccine (the Lanzhou lamb rotavirus vaccine [LLR], the Lanzhou Institute of Biological Products, Lanzhou, China) licensed in China in 2000, but this vaccine cannot be covered in the Expanded Program on Immunization due to the lack of reliable data of LLR protection on childhood RV infections. In Suzhou, consistent with the overall situation in China, oral RV vaccines have served as category II vaccine (other vaccines administered voluntarily by citizens at their own expense, compared with the first type vaccine provided free of charge by the government), and 2 RV vaccines (RotaTeq and LLR) are licensed for use.

In China, there has been a lack of studies on RV infectious diarrhea in children with large samples and accurate descriptions of the disease burden in this area. To inform policy makers on RV diarrhea prevention and promote the integration of medical treatment and prevention to reduce diarrhea attributed to RV in children, a retrospective analysis was conducted on long-time routine surveillance of children <5 years of age with RV infection in Suzhou University Affiliated Children's Hospital (SCH). This study obtained the prevalence, seasonality and strain diversity of RV infection among diarrhea children <5 years of age in Suzhou using data generated by the hospital information system and electronic medical records of SCH from 2013 to 2019.

Accepted for publication December 26, 2021

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The opinions presented in this study are those of the authors and do not necessarily represent those of Merck Sharp & Dohme Corp.

This study was sponsored by Fudan University. Funding was provided by the Shanghai New Three-year Action Plan for Public Health (Grand No. GWV-10.1-XK16) and A Merck MISP Grant (No. 59169).

The authors have no conflicts of interest to disclose.

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ISSN: 0891-3668/22/4105-0375

DOI: 10.1097/INF.00000000000003463